

Q1. Silver has FCC structure and its atomic radius is $1.414 \text{ \AA}$. Find the interplanar spacing of the (200) and (111) planes.

- **Given:** - $r = 1.414 \text{ \AA}$
- **Formula:** - $d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$ and $r = \frac{a}{2\sqrt{2}}$ for FCC structure
- **Solution:** -
 - $a = 2\sqrt{2}r = 3.99 \text{ \AA}$
 - $d_{200} = \frac{3.99}{\sqrt{2^2 + 0^2 + 0^2}} = 1.995 \text{ \AA}$
 - $d_{111} = \frac{3.99}{\sqrt{1^2 + 1^2 + 1^2}} = 2.303 \text{ \AA}$
- **Ans:** - $d_{200} = 1.995 \text{ \AA}$
- $d_{111} = 2.303 \text{ \AA}$

Q2. Cu has FCC structure. If the interplanar spacing for the set of (111) planes is $2.80\text{\AA}$, find the density and the diameter of the Cu atom. The atomic weight of Cu is 63.54.

- **Given:** - for FCC, $n=4$, $d_{111}=2.08\text{\AA}$
- $M=63.54$, $N=6.023 \times 10^{26}$ /kg-mole
- **Formula:** - $d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$, $r = \frac{a}{2\sqrt{2}}$, $\rho = \frac{nM}{Na^3}$
- **Solution:** -
 - $a = d_{111} \sqrt{1^2 + 1^2 + 1^2} = 2.08 \times 10^{-10} \times \sqrt{3} = 3.6\text{\AA}$
 - $\rho = \frac{4 \times 63.54}{6.023 \times 10^{26} \times (3.6 \times 10^{-10})^3} = 9044.547 \text{ kg/m}^3$
 - $r = \frac{a}{2\sqrt{2}} = \frac{3.6}{2\sqrt{2}} = 1.273 \text{ \AA}$
 - $\therefore d = 2.546 \text{ \AA}$
- **Ans:** - density is 9044.547 kg/m^3 and diameter is $2.546 \text{ \AA}$

Q3. If the X ray of wavelength $1.549 \text{ \AA}$ will be reflected from the crystal with interplanar spacing $4.255 \text{ \AA}$. Calculate the smallest glancing angle and the highest order of reflection that can be observe

• **Given:** - $\lambda = 1.549 \text{ \AA}$, $d = 4.255 \text{ \AA}$.

• **Formula:** - $2d \sin \theta = n \lambda$

• **Solution:** - For smallest glancing angle

•
$$n = n_{\min} = 1$$

•
$$2d \sin \theta_{\min} = \lambda$$

•
$$\sin \theta_{\min} = \frac{\lambda}{2d} = \frac{1.549}{2 \times 4.255}$$

•
$$\theta_{\min} = 10.4875^\circ$$

For Highest order, $n_{\max}$,

$$\sin \theta = 1$$

$$2d = n_{\max} \lambda$$

$$n_{\max} = \frac{2d}{\lambda} = \frac{2 \times 4.255}{1.549} = 5.49 = 5$$

Ans: - $\theta_{\min} = 10.48^\circ$

$$n_{\max} = 5$$

Q4. In comparing the wavelengths of two monochromatic X-ray lines it is found that line A gives 1st order Bragg's maximum at a glancing angle of 30° to the smooth face of a crystal. Line B of known wavelength of $0.97 \text{ \AA}$ gives 3rd order reflection maximum at a glancing angle of 60° with the same face of the crystal. Find the wavelength of line A.

- **Given:** - line A: - $n_1=1$, $\theta=30^\circ$, $\lambda_1=?$
- line A: - $n_1=3$, $\theta=60^\circ$, $\lambda_2=0.97 \text{ \AA}$
- **Formula:** - $2d\sin\theta=n\lambda$
- **Solution:** - $2d\sin\theta_1=n_1 \lambda_1$ and $2d\sin\theta_2=n_2 \lambda_2$
- $$\frac{\sin\theta_1}{\sin\theta_2} = \frac{n_1\lambda_1}{n_2\lambda_2}$$
- $$\lambda_1 = \lambda_2 \frac{\sin\theta_1}{\sin\theta_2} \cdot \frac{n_2}{n_1} = 0.97 \times 10^{-10} \frac{\sin 30}{\sin 60} \times 3$$
- $$\lambda_1 = 1.68 \text{ \AA}$$
- **Ans:** - wavelength of line A is $1.68 \text{ \AA}$

Q5.In NaCl crystal, the spacing between the successive (100) plane is $2.82 \text{ \AA}$. X-ray incident on the surface of the crystal is found to give rise to first order Bragg reflection at glancing angle 8.8° . Calculate the wavelength of X-ray.

- **Given:** - $(hkl)=(100)$, $d=2.82 \text{ \AA}$, $n=1$, $\theta=8.8^\circ$

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- **Formula:** - $2d\sin\theta=n\lambda$

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- **Solution:** - $\lambda = \frac{2d\sin\theta}{n} = 2 \times 2.82 \times 10^{-10} \times \sin 8.8^\circ$

- $\lambda = 8.628 \times 10^{-11} \text{ m}$

- $\lambda = 0.8628 \text{ \AA}$

- **Ans:** - wavelength of x-ray is $0.8628 \text{ \AA}$

Q6. Monochromatic X-ray beam of wavelength $\lambda = 5.8189 \text{ \AA}$ is reflected strongly for glancing angle of 0.7586° in the first order by certain planes of cubic crystal of lattice constant $3 \text{ \AA}$. Determine Miller indices of the possible reflecting planes.

- **Given:** $\lambda = 5.8189 \text{ \AA}$, $\theta = 75.86^\circ$, $n = 1$, $a = 3 \text{ \AA}$

- **Formula:** $2d \sin \theta = n\lambda$

- $$d = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

- **Solution:** $2d \sin \theta = n\lambda$

- $$d = \frac{n\lambda}{2 \sin \theta} = \frac{1 \times 5.8189 \times 10^{-10}}{2 \times \sin 75.86} = 3 \times 10^{-10} \text{ m}$$

- but, $d = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$

- $$\sqrt{h^2 + k^2 + l^2} = \frac{a}{d} = \frac{3 \times 10^{-10}}{3 \times 10^{-10}} = 1$$

- Hence the only possible (hkl) are (100), (010), (001)

- **Ans:** - miller indices (hkl) is (100).

Q7. Monochromatic high energy X-rays are incident on a crystal. If first order reflection is observed at an angle 3.4° at what angle would second order reflection be expected?

- **Given:** - $n_1=1$, $\theta_1=3.4^\circ$, $n_2=2$, $\theta_2=?$

- **Formula:** - $2d\sin\theta=n\lambda$

- **Solution:** - $2d\sin\theta_1=n_1\lambda$ and $2d\sin\theta_2=n_2\lambda$

- $$\frac{\sin\theta_1}{\sin\theta_2} = \frac{n_1}{n_2}$$

- $$\sin\theta_2 = \frac{n_2 \sin\theta_1}{n_1} = \frac{2}{1} \times \sin 3.4 = 0.1186$$

- $$\theta_2 = \sin^{-1} 0.1186 = 6.812^\circ$$

- **Ans:** - hence second order reflection is expected at 6.812°

Q8. Calculate the glancing angle on the (100) plane of rock salt crystal with lattice constant $2.125 \text{ \AA}$. Given the wavelength of the incident X ray is $0.592 \text{ \AA}$ and this is a case of 2nd order maximum

- **Given :** - $(kl) = (100)$, $a = 2.125 \text{ \AA}$, $n = 2$, $\lambda = 0.592 \text{ \AA}$

- **Formula:** - $d = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$; $2d \sin \theta = n \lambda$

- **Solution:** -

- $d = \frac{2.125}{\sqrt{1^2 + 0 + 0}} = 2.125 \text{ \AA}$

- $2d \sin \theta = n \lambda$

- $\sin \theta = \frac{n \lambda}{2d}$

- $\sin \theta = \frac{2 \times 0.592}{2 \times 2.125}$

- $\theta = 16.176^\circ$

- **Ans:** - The glancing angle is 16.176°

Q9. Calculate the wavelength of the X ray beam that suffers first order Bragg reflection at a glancing angle $8^{\circ}35'$ from a NaCl crystal with the lattice spacing of 0.282nm. Also calculate the maximum order of diffraction possible.

- **Given:** - $n=1$, $\theta=8^{\circ}35'$, $d=0.282\text{nm}$

- **Formula:** - $2d\sin\theta=n\lambda$

- **Solution:** -

- $$\lambda = \frac{2d\sin\theta}{n} = 2 \times 0.282 \times 10^{-9} \sin 8^{\circ}35'$$

- $$\lambda = 0.842 \text{Å}$$

- In Bragg's law, d and λ are constants. Hence, for maximum order $n_{\max}$, $\sin \theta = 1$

- $$2d = n_{\max}\lambda$$

- $$n_{\max} = \frac{2d}{\lambda} = \frac{2 \times 0.282 \times 10^{-9}}{0.842 \times 10^{-10}} = 6.7$$

- Here, since $n_{\max} < 7$, 7th order is not visible. Hence the highest order is 6

- **Ans:** - $\lambda = 0.842 \text{Å}$, $n=6$

Q10. The first order X ray diffraction occurs at a glancing angle of 9.6° from the (110) planes of a calcite crystal. If the lattice constant of the calcite is $3.036 \text{ \AA}$, calculate the wavelength of x ray.

- **Given :-** $\theta = 9.6^\circ$, $n=1$, $(hkl)=(110)$, $a=3.036 \text{ \AA}$

- **Formula: -** $d = \frac{a}{\sqrt{h^2+k^2+l^2}}$; $2d\sin\theta = n\lambda$

- **Solution: -**

- $$d = \frac{3.036}{\sqrt{1^2+1^2+0}} = 2.1468 \text{ \AA}$$

- $$\lambda = \frac{2d\sin\theta}{n} = \frac{2 \times 2.1468 \times \sin 9.6}{1}$$

- $$\lambda = 0.716 \text{ \AA}$$

- **Ans: -** The wavelength of X ray is $0.716 \text{ \AA}$

1. Find the interplanar spacing between family of planes (111) in a crystal of lattice constant $3\text{\AA}$.
2. In a crystal whose primitive vectors are $1.2\text{\AA}$, $1.8\text{\AA}$ and $2\text{\AA}$ for (231) plane that cuts an intercept of $1.2\text{\AA}$ along the x-axis, find the length of the intercepts along the y and z-axis.
3. Find the miller indices of a set of parallel planes which makes the intercepts in the ratio $3a:4b$ on the x and y axis respectively and are parallel to the z axis. The basis vectors of the unit cell are a, b, c along x, y and z axis respectively
4. Calculate the wavelength of X ray reflected from the faces of a FCC crystal with lattice constant $2.82\text{\AA}$. If the second order bragg reflection occurs at a glancing angle of 17.167° .
5. X rays of unknown wavelength give first order Bragg's reflection at glancing angle 20° with (212) planes of copper having FCC structure. Find the wavelength if lattice constant for copper is $3.615\text{\AA}$
6. The first order diffraction is found to occur at a glancing angle of 9° . Calculate the wavelength of X-ray and the glancing angle for second order diffraction if the spacing between the adjacent plane is $2.51\text{\AA}$.
7. Calculate the Bragg angle if (200) planes of a BCC crystal with lattice parameter $2.814\text{\AA}$ give second order reflection with X-rays of wavelength $0.71\text{\AA}$.
8. Calculate the maximum order of diffraction if X-rays of wavelength $0.819\text{\AA}$ incident on a crystal of lattice spacing 0.282 nm .
9. Monochromatic high energy X rays are incident on a crystal. If 1st order reflection is observed at an angle 3.4° , at what angle 2nd order reflection be expected?
10. Calculate Bragg angle if (200) planes of a BCC crystal with lattice parameter $2.814\text{\AA}$ give second order reflection with X rays of wavelength $0.71\text{\AA}$